MILITARY SPECIFICATION

INDICATOR, ATTITUDE ARU-18/A, DIRECTOR

This amendment forms a part of MIL-I-38258(USAF), dated 28 May 1965, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 Delete "first pargraph" and add: "2.1 Government documents".

* Add: "2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specificatins and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Under "SPECIFICATIONS, Federal", delete "QQ-P-416" and add:
"PPP-B-601 Boxes, Wood, Cleated Plywood".

Under "SPECIFICATIONS, Military", delete "MIL-S-6872", "MIL-C-7769", "MIL-M-26512", "MIL-R-26667", and "MIL-D-70327". Add the following:

"DOD-D-1000 Drawings, Engineering And Associated Lists
"MIL-C-83488 Coating, Aluminum, Ion Vapor Deposited"

Add boxed paragraph to bottom center of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

* Under "STANDARDS, MILITARY," delete "MS17322", "MS33586" and title. Add the following:

"MIL-STD-781 Reliability Tests Exponential Distribution
MIL-STD-785 Reliability Program For Systems And Equipment, Development And Production."

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MIL-STD-794	Part and Equipment, Procedures For Packaging And Packing Of
MIL-STD-889	Dissimilar Metals"
DOD-STD-1866	Soldering Process General (Non Electrical Metric)

* Add: "2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* Add: "2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

Delete "3.1" and add:

"3.1 First article. When specified in the contract or purhcase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4)."

3.3.4 Line 6 Delete "MS33586" and add: "MIL-STD-889."

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3.4.3 Line 4 delete "MIL-R-26667" and add: "MIL-STD-785."

3.4.3.1, 3.4.3.2, and 3.4.4: Delete these three paragraphs and add the following replcements:

3.4.3.1 Reliability. The indicator shall have a minimum Lower Test mean time between failures, MTBF, (0) of 750 hours which shall be demonstrated during reliability testing.

3.4.3.2 Longevity. The indicator shall have a minimum operating life of 5000 hours before wearout failures occur.

Delete "3.4.4."

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3.6 Line 6 Delete "MIL-D-70327" and add: "DOD-D-1000."

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3.7.7 Lines 1 and 2 delete "An MS17322-2A or -6A." Line 3 delete "or equal,".

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3.13 Line 2 and 3 delete "MIL-S-6872" and add: "DOD-STD-1866."

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3.17.2 Delete paragraph entirely and add the following:

"3.17.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practical, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the congnizant Government Engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

Add: "3.20 Recycle and reclaimed materials. The use of recycle and reclaimed materials shall be encouraged to the maximum extent possible, without jeopardizing the intended end use of the item."

Delete "4.1" and add new:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Add: "4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

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* Delete 4.2 and add:

4.2 Classification of tests. The inspection and test requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Acceptance test (see 4.5)"

Delete "4.3" and add:

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3.1 of this specification."

* 4.4 Delete "Preproduction testing" and add: "First article inspection."

4.4.2 Title and line 1: Delete "Preproduction" in both places and add: "First article tests."

4.6.37.1 Delete existing paragraph and add:

"4.6.37.1 Reliability assurance. Three indicators shall be subjected to a Reliability Qualfication Test, in accordance with MIL-STD-781, to demonstrate the MTBF requirements stated in 3.4.3.1. Test Plan XIV C shall be used. The test environment shall be in accordance with MIL-STD-781, 50.5. During each operational period and with each indicator properly connected, the following conditions shall exist:"

4.6.37.2 Delete existing paragraph and add:

"4.6.37.2 Longevity. At the completion of testing specified in 4.6.37.1 through 4.7.37.1.2 testing of one of the three indicators shall be continued for an additional 3000 hours in accordance with the test conditions stated in 4.6.37.1"

4.6.37.3 Delete existing paragraph and add:

"4.6.37.3 Acceptance. A production reliability acceptance test shall be conducted in accordance with MIL-STD-781 to demonstrate that the production indictors meet the MTBF requirements of 3.4.3.1. Sample size shall be 10% of the lot size with a minimum of three items and a maximum of 20 items for the sample size."

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4.6.44 Lines 3 and 4 delete "Appendix A to MIL-M-26512" and add: "3.4.4."

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Add new paragraph 4.7 as follows:

"4.7 Inspection of packaging. Tests for methods of preservation and packaging will be accomplished in accordance with Section 4 of MIL-P-116 to assure compliance with Section 5 of this specification."

* Delete Section 5 in its entirety and add:

"5 PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indictor shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method III of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop, tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2.)

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

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5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129."

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* 6.2b Delete.

6.2 Add "d." as follows:

"d. Indicate levels of packaging and packing required (see Section 5)."

Add new paragraph 6.4 as follows:

"6.4 First article. When a first article is required, the item will be tested. The contracting officer should include specific instructions in acquisitions documents regarding arrangements for examinations, test and approval of the first article."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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